Master's Comprehensive Examination

Part I Theory (90 minutes) March 8, 2003

Instructions: *Answer all four questions. Open book and open notes.*

1 (a). Suppose two independent random samples of patients are treated with two types of decongestant tablets for 24 hours with the following results.

Brand	Sample Size	No. of Patients Relieved
1	80	56
2	100	58

Find a 95% confidence interval for $p_1 - p_2$, the difference between the proportion of all patients who would obtain relief with Brand 1 and Brand 2.

(b) A 98% confidence interval for the mean of a normal distribution based on a sample of size 20 was found to be $[60, 83]$. What were the sample mean and variance based on the sample of 20?

2. (a) Let $X_1, X_2, \ldots, X_{400}$ be independent observations on a random variable whose mean is 2 and whose variance is 4. Find the approximate probability that the sample mean lies between 2 and 2.2. (Your final answer must be one number.)

(b) The number of errors on a page of a manuscript has a Poisson distribution with parameter $\lambda = 2$. What is the approximate probability that there are more than 934 errors in a 450 page manuscript? (Your final answer must be one number.)

3. (a) Five chips are marked 1 2 3 4 5 respectively. Two chips are drawn at random without replacement. Let Y = larger of the two numbers on the chips. Find the variance of Y.

(b) Let Z be a random variable with probability distribution

z	2	3	4
$f(z)$	.3	.3	.4

If Z_1, Z_2, Z_3 are independent and distributed as Z what is the probability that $Z_1 + Z_2 + Z_3 = 8$?